

Assignment 2

Neuromembrane Lab 2: Introduction

The purpose of this assignment is to help you visualize the process of the action potential. We will go through multiple simulations. **Be sure to answer all the questions throughout this assignment.** The questions are in red.

Goals:

1. To observe the all-or-nothing property of action potentials
2. Explore the ionic dependence of the action potential

Brief Background:

Depolarization triggers the action potential

Injection of positive current by excitatory postsynaptic potentials (EPSPs) is the usual trigger of an action potential in neurons and muscle fibers. In the action potential simulations, you will be injecting a positive current into the neuron with an electrode to trigger action potentials.

Action Potentials using Neuromembrane Simulator

<https://neuromembrane.biology.ualberta.ca/>

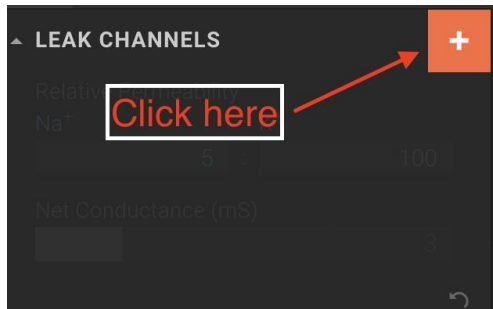
How to use the program:

1. Select “Log in as a guest” button.
2. You’ll now be inside of the action potential simulation.
 - a. **Tip:** Double-check that this is the simulation you’re in by looking at the top of the panel to the left. It should say “Action Potential Simulation.”
3. **Main page sections:**
 - **Left Section:** Allows you to adjust the parameters necessary for the simulation and is composed of several boxes that allow you to control the leak channels, Na⁺/K⁺ pump, initial membrane voltage, ion concentration settings, etc.
 - **Middle Section:** Shows a plain lipid bilayer upon which you can build your excitable membrane. Clicking the “+” sign at the top right of the box adds that particular item to the membrane in the right half of the window and allows you to adjust more detailed parameters for that item. These parameters can be adjusted by either clicking on the number, directly typing it in, or by dragging the slider left and right. The small counterclockwise arrow in the bottom right hand corner of each box allows you to revert back to default values.
 - **Right Section:** Shows the membrane simulation. The Na⁺ and K⁺ ions can be seen in the top half of the window, which represents the intracellular side. Adjusting the concentrations of the ions results in a greater (or lower) number of ions in the appropriate field of view.

Experiment 1 – The All or Nothing Principle

Steps to complete Experiment 1:

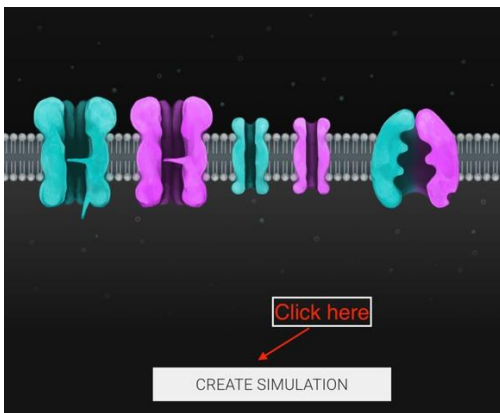
1. Let's add ion channels and the sodium-potassium pump! Within the control panel on the left-hand side, you'll see several components that you can add to your cell membrane. These components include leak channels, Na⁺ channel, K⁺ channel, and the Na⁺/K⁺ pump. **Add all these components by clicking the + button.** See screenshot below.



Tip: Once these elements have been added to the membrane, you can hover your mouse over them to see their labels.

2. Now that these elements are reflected on your cell membrane (see screenshot below), let's adjust the “external positive current” that you will be applying to the membrane through an electrode. Change the external current panel so that you only inject current ONCE. It should now look like this:

Current (μA)	Time (ms)
0	10
20	2
0	10
0	2
0	16



3. Now let's see this neuron in action! Select the “Create Simulation” button located under the cell membrane. See screenshot.

4. Familiarize yourself with the new panel. You'll now see a new panel to the left titled "Stats." This panel includes various graphs for different elements of the action potential. The first graph, titled "Voltage vs Time," will display the membrane potential of the neuron (i.e., this is the voltmeter output). The second graph, titled "Current vs Time," will display the stimuli received by the neuron. Don't worry about the last two graphs.
5. Change the playback speed to $\times\frac{1}{2}$ so that you can easily follow the changes that happen in the membrane of this neuron. Now, select the play button at the bottom of the cell membrane to begin the animation. Take note of the movement of ions before a stimulus is received, during the action potential, and after. Also note differences on the first two graphs.

a. Did you get an action potential (it must cross the threshold of "0" to be an AP)?

Circle: Yes or **No**

b. Record the max voltage using the "Voltage vs Time" graph. To do so, you will click the autoscale button  and place your cursor over the highest point the line reached and write down the first value. You can also click "Full Statistics" to expand the left panel and provide you more information, including the max voltage.

Max Voltage = **-62.753 mV**

6. Click the button to the left of the play button  to return to the settings page. You will want to increase the current of your stimulus. Return the external current panel and increase your stimulus current by 1 μA . It should now look like this:

Current (μA)	Time (ms)
0	10
21	2
0	10
0	2
0	16

7. Click "Create Simulation". Repeat "step 5" above for this new simulation.

a. Did you get an action potential? Yes or **No**

b. Record the max voltage using the "Voltage vs Time" graph. To do so, you will place your cursor over the highest point the line reached and write down the first value. You can also click "summary" to see the left panel expand and provide you more information, including the max voltage.

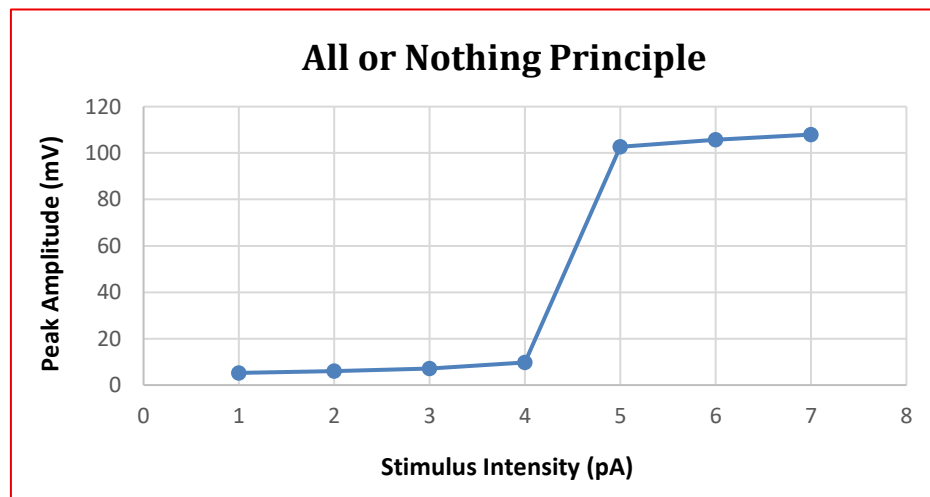
Max Voltage = **-61.961 mV**

8. You will now repeat the above steps **FIVE MORE TIMES**, increasing by 1 μA each time. Track your results in the chart below. You can find the resting membrane potential (RMP) on top of the settings page under membrane settings, listed as “GHK Membrane Potential”. Hint: The RMP shouldn’t change across these simulations but stays the same. You will then subtract the difference between the max voltage and the RMP, to get the amplitude of the peak. It should be the absolute value (in other words, it will always be a positive difference value).

Simulation #	Stimulus Intensity (μA)	Max Voltage (mV)	RMP (mV)	The absolute difference between the RMP and Max Voltage (Peak Amplitude)
1	20	-62.753	-68	5.247
2	21	-61.961	-68	6.039
3	22	-60.813	-68	7.187
4	23	-58.250	-68	9.750
5	24	34.636	-68	102.64
6	25	37.686	-68	105.69
7	26	39.939	-68	107.94

9. You will now want to input these numbers into excel to create a figure for “Stimulus Intensity vs. Peak Amplitude”. In the figure you create, stimulus intensity should be on the x-axis and Peak Amplitude should be on the y-axis. If you don’t know how to create a figure in excel, refer to the instruction sheet on our Canvas homepage.

10. Paste the figure you created in the space below:



11. Experiment summary explanation: How does this demonstrate the “all or nothing principle?”

The “All or Nothing Principle” is demonstrated by the sudden jump in the “Peak Amplitude” data (92.89 mV difference) which is seen between “Stimulus Intensity” #4 and #5 (Simulation #4 & #5). This sudden jump in the data represent the concept that the cell will only fire an action potential (signal) when the membrane voltage crosses the threshold of “0.”

Experiment 2 – Ionic Dependence of the Action Potential $[Na^+]_o$

Steps to complete Experiment 2:

1. Log out of the neuromembrane simulation and log back in as a “guest”. This will reset all your settings.
2. You will want to re-add the leak channels, Na^+ channels, K^+ channels, and the Na^+/K^+ pump to the membrane by clicking the “+” next to each component.
3. Like the experiment above, we will only want to inject one stimulus current at a time. Additionally, this current needs to be strong enough for the action potential to reach threshold. As such, change your external current panel to the following:

Current (μA)	Time (ms)
0	10
25	2
0	10
0	2
0	16

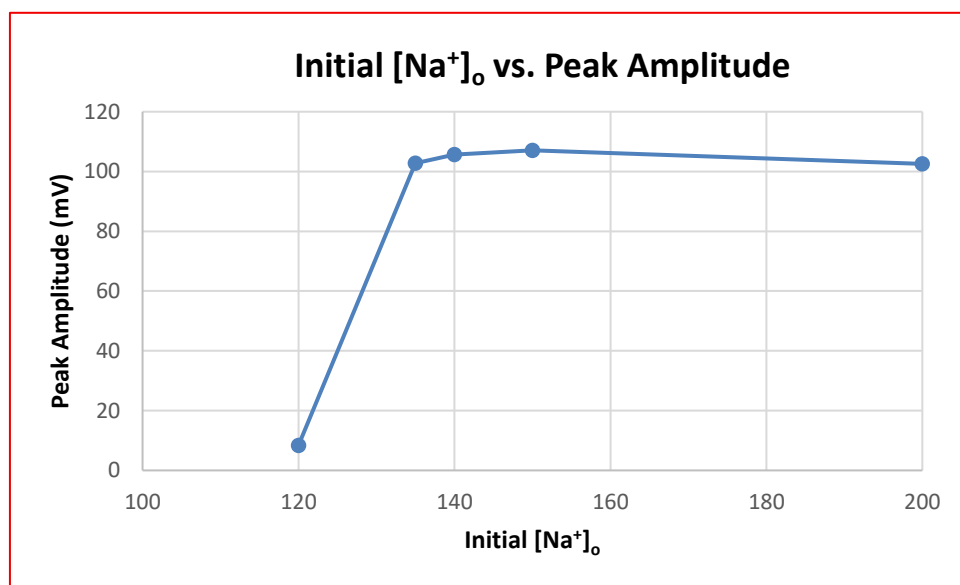
4. We will now be manipulating the concentration of Na^+ **outside** of the cell (in the extracellular space). Typically, the normal concentration of external Na^+ is 140 mM. This is the value that the simulation automatically starts at. You will run a total of FIVE simulations, only changing the external Na^+ concentration each time. To change $[Na^+]_o$, you will go to “concentration settings” and adjust the $[Na^+]$ Outside (nM) by either clicking on the number, directly typing it in, or by dragging the slider left and right.
5. Use the values listed below for your FIVE simulations. Record the same information that you recorded from the first experiment for each simulation (e.g., max voltage, RMP, and peak amplitude). Keep in mind that the RMP may now change across simulations. **Additionally, this**

time you will also be recording the time it took to reach maximum voltage. To do so, you will place your cursor over the highest point the line reached and write down the SECOND value. You can also click “summary” to see the left panel expand and provide you more information, including the max voltage and the time it occurred.

Simulation #	[Na ⁺] _o	Time to Max (ms)	Max Voltage (mV)	RMP (mV)	The absolute difference between the RMP and Max Voltage (Peak Amplitude)
1	120	11.7	-62.209	-70.6	8.391
2	135	12.5	34.102	-68.7	102.8
3	140	12.2	37.686	-68.0	105.69
4	150	11.9	40.281	-66.8	107.08
5	200	11.4	40.914	-61.6	102.51

6. You will now want to input these numbers into excel to create a figure for “[Na⁺]_o vs. Peak Amplitude”. In the figure you create, [Na⁺]_o should be on the x-axis and Peak Amplitude should be on the y-axis. If you don’t know how to create a figure in excel, refer to the instruction sheet on our Canvas homepage.

7. Paste the figure you created in the space below:



8. Experiment summary explanation: How does manipulating the concentration of Na^+ **outside** of the cell impact AP intensity (e.g., amplitude)? How does it impact the speed to maximum AP voltage? Explain why this occurs.

As you increase the concentration of Na^+ outside of the cell, the AP intensity (amplitude of signal) increases in concordance. However, this is only true up until a certain point (the maximum “Peak Amplitude” value, which was 107.08mV @ 150 $[\text{Na}^+]_o$ in this case), and then the AP intensity continues to decrease from that point on. Increasing the concentration of Na^+ outside of the cell also increases the speed (time required) to the maximum AP intensity. Both of these are due to the fact that in order to fire an AP, a minimum Na^+ threshold must be reached...the more Na^+ there is available outside the cell, the faster and stronger the AP is.

Experiment 3 – Ionic Dependence of the Action Potential $[\text{Na}^+]_i$

Steps to complete Experiment 3:

1. Log out of the neuromembrane simulation and log back in as a “guest”. This will reset all your settings.
2. You will want to re-add the leak channels, Na^+ channels, K^+ channels, and the Na^+/K^+ pump to the membrane by clicking the “+” next to each component.
3. Like the experiment above, we will only want to inject one stimulus current at a time. Additionally, this current needs to be strong enough for the action potential to reach threshold. As such, change your external current panel to the following:

Current (μA)	Time (ms)
0	10
25	2
0	10
0	2
0	16

4. We will now be manipulating the concentration of Na^+ **inside** of the cell (in the extracellular space). Typically, the normal concentration of internal Na^+ is 10 mM. This is the value that the simulation automatically starts at. You will run a total of FIVE simulations, only changing the internal Na^+ concentration each time. To change $[\text{Na}^+]_i$, you will go to “concentration settings” and

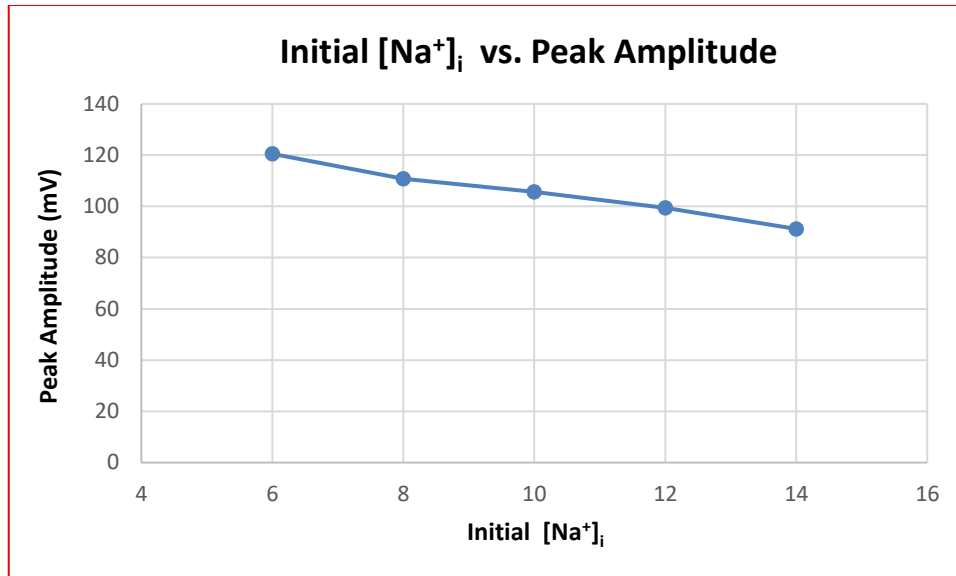
adjust the $[\text{Na}^+]_{\text{i}}$ Inside (nM) by either clicking on the number, directly typing it in, or by dragging the slider left and right.

- Use the values listed below for your FIVE simulations. Record the same information that you recorded from the first experiment for each simulation (e.g., max voltage, RMP, and peak amplitude). Keep in mind that the RMP may change now across simulations. **Additionally, this time you will also be recording the time it took to reach maximum voltage.** To do so, you will place your cursor over the highest point the line reached and write down the SECOND value. You can also click “summary” to see the left panel expand and provide you more information, including the max voltage and the time it occurred.

Simulation #	$[\text{Na}^+]_{\text{i}}$	Time to Max (ms)	Max Voltage (mV)	RMP (mV)	The absolute difference between the RMP and Max Voltage (Peak Amplitude)
1	6	11.9	52.509	-68.0	120.509
2	8	12.0	42.828	-68.0	110.828
3	10	12.2	37.686	-68.0	105.686
4	12	12.4	31.354	-68.1	99.454
5	14	12.8	23.104	-68.1	91.204

- You will now want to input these numbers into excel to create a figure for “ $[\text{Na}^+]_{\text{i}}$ vs. Peak Amplitude”. In the figure you create, $[\text{Na}^+]_{\text{i}}$ should be on the x-axis and Peak Amplitude should be on the y-axis. If you don’t know how to create a figure in excel, refer to the instruction sheet on our Canvas homepage.

- Paste the figure you created in the space below:



7. Experiment summary explanation: How does manipulating the concentration of Na^+ inside of the cell impact action potential intensity (e.g., amplitude)? How does it impact the speed to maximum AP voltage? Does it influence RMP? Explain why all this occurs.

As the initial concentration of Na^+ inside of the cell steadily increases, the AP intensity (“Peak Amplitude”) simultaneously decreases as a steady rate. In the same manner that the AP intensity is weakened when “ $[Na^+]_i$ ” increases, the speed to reach the maximum AP voltage is decreased (also can be thought of as an increase in the required amount of time). Lastly, I do not believe the “RMP” is affected by an increase in initial concentration of Na^+ inside of the cell (or at least not in the range of 6-14 mM “Initial $[Na^+]_i$ ” which we experimented with in this lab). I reached this conclusion because I don’t believe the 0.1 change in RMP was statistically significant.

Experiment 4 – Ionic Dependence of the Action Potential $[K^+]_o$

Steps to complete Experiment 4:

1. Log out of the neuromembrane simulation and log back in as a “guest”. This will reset all your settings.
2. You will want to re-add the leak channels, Na^+ channels, K^+ channels, and the Na^+/K^+ pump to the membrane by clicking the “+” next to each component.

3. Like the experiment above, we will only want to inject one stimulus current at a time. Additionally, this current needs to be strong enough for the action potential to reach threshold. As such, change your external current panel to the following:

Current (μA)	Time (ms)
0	10
25	2
0	10
0	2
0	16

4. We will now be manipulating the concentration of **K⁺ outside** of the cell (in the extracellular space). Typically, the normal concentration of external K⁺ is 4 mM. This is the value that the simulation automatically starts at. You will run a total of FIVE simulations, only changing the external K⁺ concentration each time. To change [K⁺]_o, you will go to “concentration settings” and adjust the [K⁺] Outside (nM) by either clicking on the number, directly typing it in, or by dragging the slider left and right.

10. Use the values listed below for your FIVE simulations. **Record the following information, which is different from above:** RMP, time to maximum voltage, max voltage, time to minimum voltage, min voltage, and the difference between time to max and time to min (in other words, the duration of the action potential from start to finish). Keep in mind that the RMP may change now across simulations. To find these values, you will place your cursor over the highest and lowest points on the line and write down the FIRST and SECOND values for both. You can also click “summary” to see the left panel expand and provide you more information, including the max voltage, min voltage, and the time it took to reach both. You will then subtract the difference between the two times to obtain the full time of the AP from start to finish (duration).

Simulation #	[K ⁺] _o	RMP (mV)	Time to Max (ms)	Max Voltage (mV)	Time to Min (ms)	Min Voltage (mV)	AP Duration (Time to Min – Time to Max) (ms)
1	4	-68.0	12.2	37.686	13.4	-87.631	1.2
2	5	-65.7	11.7	35.223	12.9	-82.784	1.2
3	6	-63.6	11.4	33.682	12.7	-78.636	1.3
4	7	-61.6	11.3	28.590	12.6	-75.009	1.3
5	8	-59.8	11.2	21.374	12.6	-71.755	1.4

11. **Experiment summary explanation:** How does manipulating the concentration of **K⁺ outside** of the cell impact the following features: Be sure to also explain WHY.

- a. Resting membrane potential (RMP):

The resting membrane potential is increasing (becoming more depolarized) because the RMP is dependent on both the electrical charge outside as well as inside the cell. Since we are increasing the initial concentration of K^+ outside of the cell, this lowers the amount of K^+ which needs to leave the cell to establish equilibrium (less K^+ ions end up leaving the cell). As a result, the inside of the cell is less negative and therefore the RMP decreases (as the difference in charge inside and outside the cell is now reduced in its magnitude)

- b. Max voltage (i.e., height of the AP):

The max voltage is directly dependent on how much Na^+ enters the cell once the “voltage-gated Na^+ channels” open up. Since an increase in the initial concentration of K^+ outside of the cell leads to a greater charge (less negative) inside the cell, the amount of Na^+ which needs to enter the cell is decreased. Therefore, less Na^+ = AP intensity or “max voltage” is decreased.

- c. Time to max voltage (i.e., how quickly the AP fires):

The reason that the “time to max voltage” decreases (and the reason why the AP fires quicker) when there is an increased initial concentration of K^+ outside of the cell is, again, due to the RMP being increased. Since the RMP is closer to the threshold of “0,” it takes less work (and therefore less time is required for the work to take place) to reach the threshold and then fire an AP.

- d. Min voltage (i.e., the hyperpolarization value):

Since more K^+ is present outside the cell membrane initially, and there is a reduction in the number of the K^+ ions which need enter the cell, this also means that less K^+ will exit the cell membrane after firing the AP. Since less K^+ leaves the cell, the cell also experiences a decrease in the “minimum voltage” since the inside of the cell is less negative (and the RMP is increased)

- e. The duration of the AP (how long it lasts from start to finish):

Increasing the initial concentration of K^+ outside of the cell results in an increase of the AP duration. This is due to the fact that since there is a reduced outward K^+ gradient, the repolarization occurs slower and therefore the signal of AP lasts longer.

